

Section 08 — Procedures & Functions

Create Procedure

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE SP_ADD_EMPLOYEE (
    p_fname      IN VARCHAR2,
    p_lname      IN VARCHAR2,
    p_gender     IN CHAR,
    p_age        IN NUMBER,
    p_contact_add IN VARCHAR2,
    p_emp_email   IN VARCHAR2,
    p_emp_pass    IN VARCHAR2,
    p_job_ID     IN NUMBER,
    p_salary_ID  IN NUMBER
)
AS
    v_email_count NUMBER;
BEGIN
    SELECT COUNT(*)
    INTO v_email_count
    FROM EMPLOYEE
    WHERE emp_email = p_emp_email;

    IF v_email_count > 0 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Email already exists.');
```

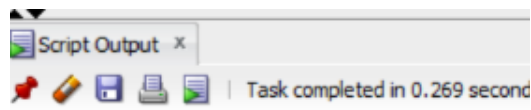
END IF;

```
    INSERT INTO EMPLOYEE
    (emp_ID, fname, lname, gender, age, contact_add,
     emp_email, emp_pass, job_ID, salary_ID)
    VALUES
    (EMP_SEQ.NEXTVAL, p_fname, p_lname, p_gender, p_age,
     p_contact_add, p_emp_email, p_emp_pass,
     p_job_ID, p_salary_ID);

    DBMS_OUTPUT.PUT_LINE('Employee added successfully.');
```

END;

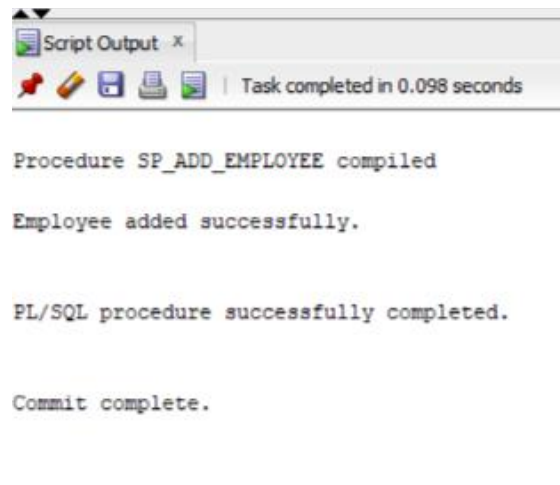
/



Procedure SP_ADD_EMPLOYEE compiled

Call Procedure

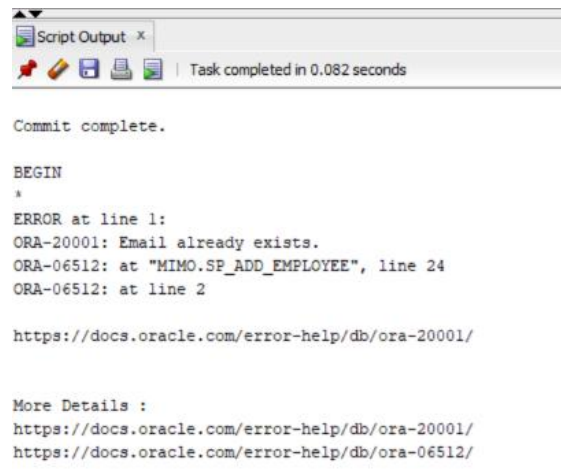
```
BEGIN
  SP_ADD_EMPLOYEE(
    'Yusuf',
    'Alshukaili',
    'M',
    29,
    'Muscat',
    'yusuf@ems.com',
    'pass123',
    1,
    1
  );
END;
/
COMMIT;
```



Call Procedure Again

```
BEGIN
  SP_ADD_EMPLOYEE (
    'Yusuf',
    'Alshukaili',
    'M',
    29,
    'Muscat',
    'yusuf@ems.com',
    'pass123',
    1,
    1
  );
END;
/
```

ORA-20001 raised because the email yusuf@ems.com already exists in the table



The screenshot shows the 'Script Output' window in SQL Developer. The status bar at the top indicates 'Task completed in 0.082 seconds'. The output text shows a successful commit followed by an error at line 1: 'ORA-20001: Email already exists.' The error also points to line 24 of 'MIMO.SP_ADD_EMPLOYEE' and line 2 of the script. It includes links to Oracle error documentation for ORA-20001 and ORA-06512.

```
Commit complete.

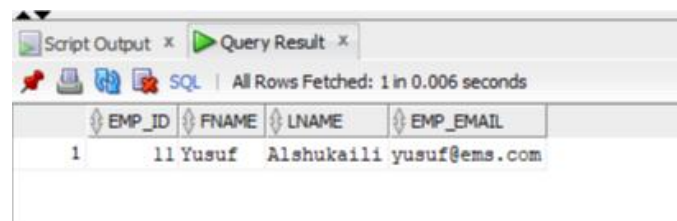
BEGIN
*
ERROR at line 1:
ORA-20001: Email already exists.
ORA-06512: at "MIMO.SP_ADD_EMPLOYEE", line 24
ORA-06512: at line 2

https://docs.oracle.com/error-help/db/ora-20001/

More Details :
https://docs.oracle.com/error-help/db/ora-20001/
https://docs.oracle.com/error-help/db/ora-06512/
```

Verify Insert | SELECT Employee

```
SELECT emp_ID, fname, lname, emp_email
FROM EMPLOYEE
WHERE emp_email = 'yusuf@ems.com';
```

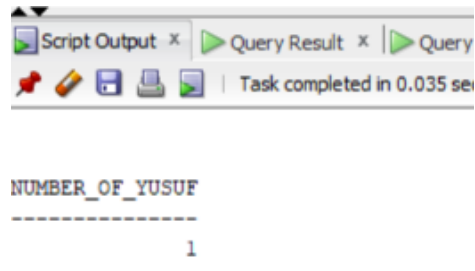


The screenshot shows the 'Query Result' window in SQL Developer. The status bar indicates 'All Rows Fetched: 1 in 0.006 seconds'. The table has four columns: EMP_ID, FNAME, LNAME, and EMP_EMAIL. One row is displayed with the values 1, ll Yusuf, Alshukaili, and yusuf@ems.com.

EMP_ID	FNAME	LNAME	EMP_EMAIL
1	ll Yusuf	Alshukaili	yusuf@ems.com

Yusuf Was Added Only Once

```
SELECT COUNT(*) AS number_of_yusuf
FROM EMPLOYEE
WHERE emp_email = 'yusuf@ems.com';
```



The screenshot shows a SQL query result window with the following details:

- Tab: Query Result x
- Status: Task completed in 0.035 seconds
- Column: NUMBER_OF_YUSUF
- Row 1: 1

NUMBER_OF_YUSUF
1

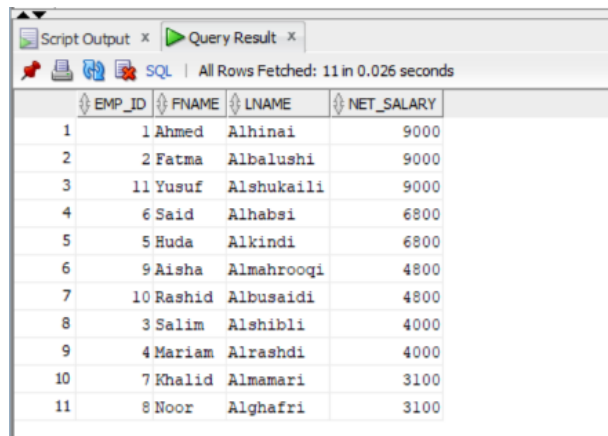
Task 2 — Create Function

```
CREATE OR REPLACE FUNCTION FN_NET_SALARY (
    p_emp_ID IN NUMBER
)
RETURN NUMBER
AS
    v_net_salary NUMBER;
BEGIN
    SELECT sb.amount + sb.bonus
    INTO v_net_salary
    FROM EMPLOYEE e
    JOIN SALARY_BONUS sb
    ON e.salary_ID = sb.salary_ID
    WHERE e.emp_ID = p_emp_ID;

    RETURN v_net_salary;
END;
/
```

Call Function

```
SELECT e.emp_ID,  
       e.fname,  
       e.lname,  
       FN_NET_SALARY(e.emp_ID) AS net_salary  
FROM EMPLOYEE e  
ORDER BY net_salary DESC;
```



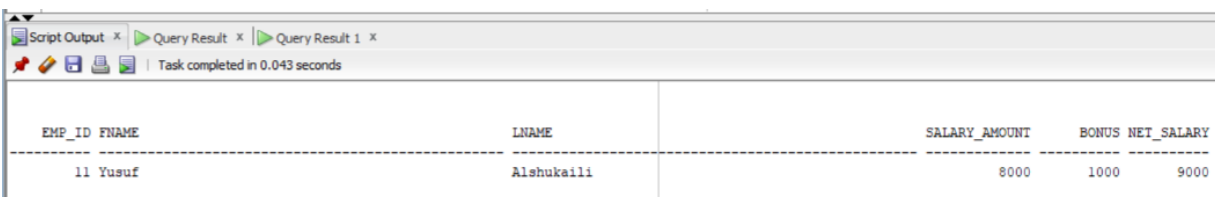
Script Output x Query Result x
SQL | All Rows Fetched: 11 in 0.026 seconds

EMP_ID	FNAME	LNAME	NET_SALARY
1	Ahmed	Alhinai	9000
2	Fatma	Albalushi	9000
3	11 Yusuf	Alshukaili	9000
4	6 Said	Alhabsi	6800
5	5 Huda	Alkindi	6800
6	9 Aisha	Almahroogi	4800
7	10 Rashid	Albusaidi	4800
8	3 Salim	Alshibli	4000
9	4 Mariam	Alrashdi	4000
10	7 Khalid	Almamari	3100
11	8 Noor	Alghafri	3100

Verify

Proves FN_NET_SALARY returns amount + bonus correctly for the newly added employee

```
SELECT e.emp_ID,  
       e.fname,  
       e.lname,  
       sb.amount AS salary_amount,  
       sb.bonus,  
       FN_NET_SALARY(e.emp_ID) AS net_salary  
FROM EMPLOYEE e  
JOIN SALARY_BONUS sb  
ON e.salary_ID = sb.salary_ID  
WHERE e.emp_email = 'yusuf@ems.com';
```



Script Output x Query Result x Query Result 1 x
Task completed in 0.043 seconds

EMP_ID	FNAME	LNAME	SALARY_AMOUNT	BONUS	NET_SALARY
11	Yusuf	Alshukaili	8000	1000	9000